



RIBA PART 3
EXAMINATION IN PROFESSIONAL PRACTICE

**September 2014
Practice Problems**

Day 1

Questions 1-5

NB – Change to scenario- for ease of differentiation, 4-classroom extension now reads as 4-classroom refurbishment.

SCENARIO

The name NDG Architects Ltd is listed in the current RIBA Directory of UK Chartered Practices.

Established in 1998, the practice has a wide range of design and development experience across a number of private and public sector areas, including commercial, retail, industrial, education and health care.

Exclusively based in the North West of England, NDG Architects' office is located in a converted mill building in Central Madchester.

The practice has noticed a sustained upturn in its workload, particularly educational projects, over the past year which has allowed it to forecast the next 6 months with renewed confidence.

As a candidate for the September 2014 examination you are required to assume that you are a salaried employee of NDG Architects and have been engaged at this practice for a period of 20 months whilst preparing to sit the Part 3 examination.

Companies House records show that NDG Architects Ltd has two directors and two associate directors.

James Noble had indicated that it was his intention to retire in 2015 and during that time to gradually cut down his direct involvement with the projects. As of 1st September 2013 he has reduced his working week to 4 days. Recently however, he has suffered some health problems, has been away from the office for an extended period and has now decided to retire with immediate effect, but remain as a consultant.

Belinda Carter architect and director. Her supplementary role within the company is to maintain the marketing operations, which prior to 2011 had generated significant new multi million pound commissions for blue chip clients. Belinda has now assumed the role of MD but has made it clear that over the next year she feels that significant restructuring will have to take place within the practice.

Michael Hart associate director, RIBA specialist conservation architect and listed on the Diocese schedule of approved architects. His reputation in this field has previously resulted in the practice being commissioned to carry out larger conservation projects attracting lottery funding. Michael is also listed as a Client Design Advisor on the RIBA's CDA register.

Sue Allen, architect and associate director is an accredited sustainability expert and has recently completed a MSc course in Advanced Sustainability at the University of Madchester. Sue also acts as the company's quality manager, responsible for the ongoing development and maintenance of the quality assurance system and manual.

Certain significant projects currently and previously handled by the office:

Proposed refurbishment & conversion of existing stables into Farm Shop and Cafe at Tally Ho Manor.

The practice has been involved with this project for a number of years, master-planning the site and completing refurbishment works at the Manor house. The current phase, the Farm Shop and Cafe project, has been on hold due to funding issues but these have recently been resolved and the project is to go ahead immediately. Planning has been secured. Approx value £600,000.

Madchester Academy

The practice has undertaken work for many years for Madchester Academy, the most recent appointment being a £2m community performing arts centre which is currently at the briefing stage.

Christchurch School, Madchester

The **4-classroom refurbishment** has recently been handed over and is currently in the rectification period. The project architect left the practice the week after practical completion and you are the only other member of staff who has worked on it recently. The School were delighted with the final project but this has been tarnished by a recent accident when one of the children lost part of a finger when it became trapped in a door jamb – the cause and liability are currently being investigated.

Recently the Board of Governors has decided to procure future projects under design and build contracts and you have been approached by a contractor with whom you have worked for many years to submit a bid for a new 6-classroom extension.

Luxury Apartments in Madchester

The practice has completed work up to Stage 3 on the £3m refurbishment of the property in Madchester into luxury apartments – work has currently been put on hold.

Potential new Housing Development for Iffy Developments

This is a potential new client who is proposing to develop the former bus depot site to provide 15 new houses. The practice are anxious to work with this developer as it could lead to a number of new commissions in the future. The project is still on hold.

Richard III College

Following the practice's extensive work at Madchester Academy, they have been appointed as architects for the library refurbishment at Richard III College, a building of some architectural merit.

The project is currently on site under a JCT Standard Building Contract with Quantities 2011.

Community Library

Madchester Council has recently closed one of its public libraries and an Action Group has been formed by the local community to try to lease the building and to run it as a library and community centre, using volunteers.

The 400 sq m, single storey building, is Grade II Listed and is over 100 years old. The practice has been approached to act as consultants.

City Gallery

The Practice has been approached to look into the viability of developing a rooftop extension to the existing gallery which is located in the centre of Madchester.

Refurbishment of Victorian house for Mr and Mrs Landers

The project was completed in February and following the end of the rectification period, the final certificate has been issued. The practice has been notified of a problem with the wood burning stove/flue.

New barn conversion for Mr and Mrs Oldsworth

This is a new commission to convert the barn into a new sustainable home.

Staff and salaries/director remuneration

James T. Noble	Director	£ 52,000.00
Belinda C. Carter	Director	£ 52,000.00
Michael Hart	Associate Director	£ 38,000.00
Sue Allen	Associate Director	£ 38,000.00
Dareem Patel	Architect	£ 35,000.00
Sven Berg	Architect	£ 35,000.00
Tomas Hjeltnes	Architect	£ 31,000.00
Renata Wang	Architect	£ 31,000.00
George Gee	Architectural Technologist	£ 27,000.00
Candidate	Architectural Assistant	£ 24,000.00
Olga Poulcek	Architectural Assistant	£ 24,000.00
Mark Jones	Architectural Technician	£ 21,000.00
Liv Grosfjelt	Year out student	£ 15,000.00
Sara Smitt	Year out student	£ 15,000.00
Sarah Jane Key	Administrative assistant	
Sophie Bright	Receptionist/ Clerical assistant	
Jane Dudley	Clerical assistant	
Anne Holding	Accounts assistant	

NB All of the characters, locations and projects of the scenario and the practice problems are entirely fictitious and any similarity or resemblance to persons, places, projects or practices is totally coincidental.

End

QUESTION 1
Page 1 of 1

Email@

Date: 3rd September 2014

To: Candidate@ndgarchitects.com

Cc:

From: MichaelHart@ndgarchitects.com

Re: New 6-classroom extension to Christchurch School, Madchester

We received a call from Mr Henratty from Roland Construction yesterday, you will recall that they have done quite a bit of work with us over the years and are currently bidding for the 6 classroom extension at Christchurch School, Madchester on a D&B basis. He asked whether we would be interested in teaming up as their architects.

I did however wake up in the middle of the night wondering whether it was the right thing to do given our ongoing appointment with the Board of Governors and the recent accident at the school.

It is a dilemma and I would appreciate your thoughts.

Could you set out your understanding of the issues with particular reference to the Professional Codes of Conduct.

If we were to team up with Roland Construction what action should we take to ensure our professional integrity?

Kind regards

Michael Hart
Associate Director

ndg architects ltd

Email@

Date: 3rd September 2014

To: Candidate@ndgarchitects.com

Cc:

From: BelindaCarter@ndgarchitects.com

Re: Community Performing Arts Centre

I have to make a presentation to School Governors at Madchester Academy where we are just about to start the briefing and design development for their £2m community Performing Arts Centre. This project is a stand-alone building which can be easily accessed out of hours. It will have sprung dance floors and changing facilities that will also be used by local community groups.

The topic of my presentation is "Stakeholders" and the Governors want to know who we think the stakeholders are on this project. They have asked how we intend to communicate with stakeholders when putting the brief together, and also how we propose to feed back to ensure everyone who matters feels they are involved with the process.

Can you please provide me with a crib sheet in bullet point format that I can refer to, as well as a rough programme that shows how things pan out up to RIBA stage 3

I do NOT want a download from the internet, but your ideas about specifically WHO we should consult and what will be the best method to communicate with the different people and groups. Don't forget, putting the brief together requires us to consult on practical issues such as maintenance, servicing and security.

Remember that, at the end of the day, it is the School that is footing the bill and so they will have the final say in what happens and how the information is fed back to stakeholders

Kind regards

Belinda Carter
Director

ndg architects ltd

QUESTION 3
Page 1 of 2

Email@

Date: 3rd September 2014

To: Candidate@ndgarchitects.com

Cc:

From: MichaelHart@ndgarchitects.com

Re: 4-classroom refurbishment

While you were away on your part 3 course yesterday we received a call from the Head Teacher, Janet Jones, at Christchurch School Madchester where, just last week, we handed over the new 4-classroom refurbishment. It seems one of the children has lost part of their finger when it became trapped in a door jamb.

In your absence I have checked the files. As you know, this a traditionally procured job where we are both lead consultants and contract administrators. It seems that whilst finger guards were specified to all doors in our door schedule, at the time of the handover they were not fitted.

It appears that you took the view that this was a minor item and not something that should hold up handover especially since the school needed to take possession of the new area without delay. The records show that we have issued a practical completion certificate with a list of outstanding items, including finger guards.

The head explained that she had also spoken to the CDM-C because she assumed that this is a health and safety matter and was told that the health and safety file was currently being compiled by the general contractor and should be available "in a couple of weeks".

This is obviously a rather sad and serious matter which I will discuss in the Directors meeting at lunch time. Could you please prepare a memo for my reference in the meeting which covers the following:

1. On what basis, at practical completion, we considered the finger guards to be a "minor item" which need not hold up handover?
2. What actions should the Practice now take over this incident to prepare ourselves for what might happen next?
3. What are likely to be the repercussions (for all parties) with the HSE.
4. Where does the CDM - C stand in all this. Does it matter that the health and safety file has not been handed to the Client (of course the CDM-C was not actually at the handover, but did receive a copy of the PC certificate and attachments)

QUESTION 3
Page 2 of 2

5. Finally, with hindsight, what could we have done at practical completion, given the pressure we were under to handover the new classrooms?

Also, can you please draft a diplomatic holding letter to the Head Teacher. Remember that although she is our contact, the actual Client is the Board of Governors.

Kind regards

Michael Hart
Associate Director

ndg architects ltd

QUESTION 4
Page 1 of 1

Email@

Date: 3rd September 2014

To: Candidate@ndgarchitects.com

From: MichaelHart@ndgarchitects.com

Re: 6-classroom extension at Christchurch School, Madchester

You will recall from my earlier email that Roland are bidding for a Design and Build 6 classroom extension on a flat site near Madchester which has a construction value of around £600,000 and would like us to team up with them as their architects.

The design has already received planning permission with some conditions on materials attached and the design has been taken to RIBA stage 3 (but no Building Regs submission)

I have already agreed a fee for production of the basic working drawings they will need to build the scheme, but I am concerned that we will spend a lot of time on other things that we might not get paid for unless we set out carefully what we are doing and what we are not doing.

Could you please do two things for me:

1. Draft an email to Mr Henratty which needs to sound positive and helpful, but sets out a fee and services proposal and states what we will do and what we will not do (to avoid disputes later)
2. Draft a fee calculation (excluding the working drawings) based upon the time you think we will need to perform the tasks using some sort of spreadsheet. Assume charge out rates at double the salary costs (don't forget N.I.)

Remember, we are not the contract administrators, but the contractor will almost certainly need professional help with things throughout the job.

I know you will have to make lots of assumptions; I want you to think carefully through the stages and processes we have to get through from first principles to get the building built and handed over. Assume a 20-week construction programme with a 4-week lead in.

Kind regards

Michael Hart
Associate Director

ndg architects ltd

QUESTION 5
Page 1 of 3

Email@

Date: 3rd September 2014

To: Candidate@ndgarchitects.com

Cc:

From: SvenBerg@ndgarchitects.com

Re: Richard III College

We have received a phone call from a distraught contractor late last night.

The £1,200,000 college library refurbishment is due to be completed in a few days time and the new 25,000 litre water booster tank in the roof void has suffered a catastrophic failure and dropped all its water on the reading rooms below.

Everyone from the contractor's and client's team have spent most of the night cleaning the mess but it is obvious that there is significant water damage to new partitions, ceilings, carpets, electrics, etc and some of the completed work will have to be stripped and replaced. There is also some damage to the existing lath and plaster ceilings and large decorative horse hair cornices.

The contractor is extremely upset as he has been doing regular work with the college for many years and this may result in him losing future work. He is also very concerned that the problem was caused as a result of the tank not being installed properly and that he will have to fund all the repairs plus the cost of liquidated and ascertained damages. He also informs me that the tank has been installed by his son's company.

Even though the construction industry is emerging from a deep recession, the costs of this could put the contractor's company out of business.

The JCT Standard Building Contract with Quantities 2011 has been used and work had been progressing nicely on site. The contractor has been very good at managing the work and the finishes are to a very high standard but they are not very good with contracts and paperwork. I don't think that we have ever had a claim from this contractor so I am not expecting any letters soon.

Our client however is very angry and I attach a letter from the college.

Can you please draft me a memo with your thoughts on who you think is responsible for this problem and what needs to happen so that the works can be completed and handed over to the college.

QUESTION 5
Page 2 of 3

Can you also please draft a carefully worded letter to our client informing them of the current contractual position.

Kind Regards

Sven Berg
Architect

ndg architects ltd

Richard III College

**St Peters Way
Roachdale
OL3 2RT**

**NDG Architects Ltd
Wharf Mill
Paccadilly
Madchester
MAD PPA**

3rd September 2014

Dear Mr Berg

You will be aware of the recent water damage caused to the newly refurbished library which was due to be open to students soon. This damage is due to faulty workmanship and we are expecting the contractor to rectify this completely at their own cost, including the existing ceilings and cornices. The college will also exercise their right under the building contract to claim liquidated and ascertained damages from the contractor for the full duration of any delay caused as a result of this.

I also instruct you as our architect and contract administrator to obtain the contractors proposal and programme for completing the work as quickly as possible.

Yours sincerely

Mr Earl Rivers

PS

I am also very concerned that such a large tank of water has been positioned over a sensitive area and should the tank room have a waterproof floor to catch any spills?

END OF DAY ONE

**Please ensure that you hand a copy of your draft answers
to your exam supervisor by 5.30pm**



RIBA PART 3
EXAMINATION IN PROFESSIONAL PRACTICE

September 2014
Practice Problems

Day 2

Questions 6-10

QUESTION 6

Page 1 of 1

Email@

Date: 4th September 2014

To: Candidate@ndgarchitects.com

From: BelindaCarter@ndgarchitects.com

Re: House refurbishment for Mr & Mrs Laanders

The office has received a complaint from a client, Harry Laanders, about a project you worked on when you had just joined the office. The project architect who ran the project as you know has left and so you are the only one that knows anything about it. You will no doubt remember the project but from my recollection it was a refurbishment of a small Victorian house. The office led the project and provided full services including contract administration of the IC 11 building contract. As part of the works the project architect designed a new fireplace, flue and associated works for a decorative wood-burning stove that the client had bought on ebay and which was installed by a specialist during the works as an ex-contract item. I understand Harry sent the specification to us which gave the heat output, flue length etc. and asked us to design the work to suit the specification.

The project was finished on time in February, a Completion Certificate was issued by Building Control and the rectification period ended in mid-August. There were no defects and I believe you issued the Final Certificate two weeks ago. Last weekend Harry decided to use the stove for the first time but the ground floor filled with smoke and he had to put it out. The installer was called and he said there was nothing wrong with the stove but that the flue was too short, the hearth and adequate ventilation were also missing. Harry called the local authority and Building Control said that they knew nothing about the stove as only the fireplace and the flue were shown on the drawings.

Harry says that the office failed in its duty to design the installation correctly, to notify Building Control and to supervise the works and that he is going to complain to the RIBA and the ARB and make a claim against the practice for the costs of rectifying the problem with the flue and ventilation so that it works safely.

As you were involved in the project can you draft a short report setting out the key issues and whether or not the practice is at fault.

Kind regards

Belinda Carter
Director

ndg architects ltd

QUESTION 7

Page 1 of 1

Email@

Date: 4th September 2014

To: Candidate@ndgarchitects.com

Cc:

From: Michael Hart@ndgarchitects.com

Re: Barn conversion for Mr & Mrs Oldsworth

We have been approached by new clients, Mr and Mrs Oldsworth, who have just bought a dilapidated barn for conversion into a new sustainable family home. The barn is located in the Peak District National Park. They have never used an architect before or carried out any building projects and they want the office to design and manage the whole process. I would like you to attend the first client meeting at which I would like you to explain the professional services that the practice can provide.

I believe that you recently attended a lecture on the New Plan of Work and so can you:

1. Using the Plan of Work website (www.ribaplanofwork.com) prepare a draft Plan showing the services that the office should provide for this type of project, the procurement route, specify the role of the architect and other members of the design team;
2. Set out the detailed services you would provide at Stages 0 and 1 including any likely additional services that you think may be required.

Kind regards

Michael Hart

Associate Director

ndg architects ltd

QUESTION 8

Page 1 of 1

Email@

Date: 4th September 2014

To: Candidate@ndgarchitects.com

From: MichaelHart@ndgarchitects.com

Re: Listed Library in Madchester

Madchester Council has recently closed one of its public libraries and an Action Group has been formed from the local community to try to lease the building and to run it as a library and community centre, using volunteers.

The 400 sq m, single storey building, is Grade II Listed and is over 100 years old.

The Action Group has been deterred by the Council from taking on the liabilities of such a Listed Building, arguing that they may not have the financial or professional resources necessary to manage, adapt and maintain the historic building. The Council has also advised of the risks of prosecution if works are undertaken without consent. This has naturally worried the Action Group members, who have come to us for advice.

The only new works which the Action Group intend to carry out comprise the construction of a public toilet and baby-change facility within the building, plus a small kitchen for tea and coffee. This, they feel, would neatly fit in an under-used storeroom at the rear of the building.

The Action Group would also need to carry out maintenance of the building including resolving issues relating to lead flashings which have been stolen from the roof.

The Action Group have approached us for advice on:

- a) How they can discover what elements of the building are listed
- b) What consents they will require for the new toilet and kitchenette
- c) The process and documentation necessary to seek the necessary consent
- d) The likely timescale to obtain the necessary consents.
- e) Whether they require consent to carry out repairs to the roof flashings

Can you draft me a short briefing note in response to these five points which I can discuss with the Action Group when I meet them next week.

Michael Hart
Associate Director

ndg architects ltd

QUESTION 9
Page 1 of 1

Email@

Date: 4th September 2014

To: Candidate@ndgarchitects.com

Cc:

From: BelindaCarter@ndgarchitects.com

Re: Proposed refurbishment & conversion of existing stables into Farm Shop and Cafe at Tally Ho Manor.

As you may recall, this project for Lord Avafewbob has been on hold for about a year, since we obtained planning approval. The client has now secured funding and wants to move the project forward and be on site within 4 months. As I am sure you are aware, since Christmas we have been at full stretch and have struggled to resource projects.

Although we need to resource Lord Avafewbob's project I am cautious about taking on a permanent member of staff as cash flow is still tight. Therefore, a short-term solution may be to employ a Part I Architectural Assistant under the supervision of an Associate.

It has been a long time since we have employed additional staff, and as you are currently completing your part III, I thought it would be good for you to evaluate whether a Part I Assistant is the most appropriate and effective means of resourcing this project. If necessary, please suggest alternatives taking into account our current financial position.

In addition, I would like you to suggest how we might find the right person for the job including drafting an advertisement suitable for industry publications such as the AJ or BD.

For the interviews, please put together an interview checklist and finally please suggest appropriate contractual arrangements for their employment?

Kind regards

Belinda Carter
Director

ndg architects ltd

QUESTION 10

Page 1 of 1

Email@

Date: 4th September 2014

To: Candidate@ndgarchitects.com

Cc:

From: JamesNoble@ndgarchitects.com

Re: City Gallery Extension

Just before I left for Majorca, I was at the city gallery for the awards presentations with Sir Jack Daw, the archive director.

He was telling me that their new project is to create a space for an exhibition of rare ancient documents and he was going to contact us next week to ask us to look into the viability of the project.

He has an open mind as to how the new gallery can be achieved but his over-riding consideration is that the exhibition opens in early summer next year.

I know the building well, as I am sure that you do too, and he confirmed my suspicions that there is no space within the existing building for this exhibition. Given the constraints of the site the only option that I can see would be to extend at roof level.

At this point the Chief Planning Officer happened to join our conversation and so I took the opportunity to get his initial reaction to a rooftop extension and it was surprisingly positive.

It is not going to be an easy project given the difficulties of the constraints of the site, access off the main road, lack of storage to say nothing of keeping the existing gallery open but it would be a high profile project for the practice and I would like to demonstrate an innovative approach to both construction and procurement.

Could you let me have your views and considered options for the best way forward to achieve their aims in terms of innovative and collaborative approaches to project management and procurement. You might also consider the key players involved and what building contracts might be appropriate, listing the advantages and disadvantages of the different approaches – I will be meeting the committee on my return and would like to put forward constructive suggestions to them.

Kind regards

James Noble
Director

ndg architects ltd

END OF DAY TWO

**Please ensure that you hand a copy of your draft answers
to your exam supervisor by 5.30pm**

DAY THREE

CHECK LIST – DO NOT FORGET TO SEND

DUPLICATE COPIES OF

C.V.

SELF EVALUATION

PEDR SHEETS

CASE STUDY

AND RESPONSES TO PRACTICE PROBLEMS

CD OF SUBMISSION (this can be sent by 8th September 2014 if necessary)

TOGETHER WITH

**DECLARATION – signed by yourself and the Exam Supervisor
Sealed draft copies of each day's responses**

**REMEMBER to obtain a Certificate of Posting and keep in a safe place to
present at interview.**